CbcSolver: Class Architecture Overview

coin-or/Cbc — next branch — src/CbcSolver.hpp / src/CbcSolver.cpp

What is CbcSolver?

CbcSolver is the top-level driver class for the Cbc MIP solver executable and library entry point, encapsulating the entire solve pipeline — parameter initialization, command-queue parsing, LP relaxation solves, preprocessing, cut/heuristic configuration, branch-and-bound search, post-processing, and result reporting — that used to live in the free functions CbcMain0/CbcMain1, now thin backward-compatible wrappers delegating to initialize() and run().

Historically, CbcSolver::run() was a single ∼14,000-line function built around one giant switch on the current command. A series of refactors (most recently: removing dead #ifndef JJF_ONE / #if 0 blocks, splitting the branch-and-bound case into six cooperating private methods, and deleting the legacy hard-coded -miplib/-unitTest self-test harness) progressively extracted coherent phases into named methods listed below, while run() itself was reduced to a slim command dispatcher. Local variables that needed to survive across these phases were promoted to private data members (see §4).

1 Public API surface

- **Construction & lifecycle:** default / from-solver / from-model / copy constructors, operator=, destructor (lines 1927–2341); initialize() sets up signal handlers and default parameters (replaces CbcMain0).
- **High-level solve:** solve(file), importModel(file), solveLp(method) — convenience one-call wrappers that chain initialize() + run() or solve the LP relaxation directly.
- **Result accessors:** status(), objectiveValue(), bestBound(), bestSolution(), numCols(), nodeCount(), iterationCount(), hasSolution() — valid after solve()/run() returns.
- **Run (replaces CbcMain1):** run(argc,argv,...) delegates to run(deque,...), the primary command-queue–driven entry point and the root of the pipeline in §2.
- **Heuristic / cut-generator configuration:** configureHeuristics(), configureCutGenerators() — public wrappers exposing internal BAB setup so callers can invoke it directly on a model without going through run().
- **User extensions:** addUserFunction(), setUserCallBack(), addCutGenerator() register CbcUser / CbcStopNow / CglCutGenerator plug-ins.
- **Accessors:** model(), babModel(), parameters(), intValue() / setIntValue(), doubleValue() / setDoubleValue(), originalSolver(), originalCoinModel(), getStatistics(), and similar simple getters/setters over the state in §4.

2 The branch-and-bound pipeline

run() dispatches on the current CbcParam command. Besides simple one-shot actions (§3), the BAB command drives the solver's core pipeline, implemented as six cooperating private methods called in sequence (see the full diagram on the next page):

1. babSetupAndRootLp() — cutoff sign flip, legacy OsiSolverLink path, solves the root LP via solveInitialLp() → applyLpMethod().
2. babConfigureBabModel() — (re)builds babModel_ from model_, configures the conflict-graph branching ranker (RankerConfig), tunes factorization frequency.
3. preprocess() — runs CglPreProcess when preprocessing is enabled.
4. babPostPreprocessCleanup() — cgraph/clique-strengthening cleanup, post-preprocess bound tightening.
5. babConfigureSearchModel() — knapsack expansion, cost-based priorities, and invokes configureHeuristics() / configureCutGenerators().
6. babExecuteSearchAndPostprocess() — SOS/priority setup, the actual babModel_->branchAndBound() call, search-statistics bookkeeping, and finally postprocess() to translate the solution back and report results.

Each phase returns a status code consumed by run() to decide whether to continue, break the BAB case, or return from run() outright — preserving the exact control flow of the original monolithic function.

3 Other run() actions

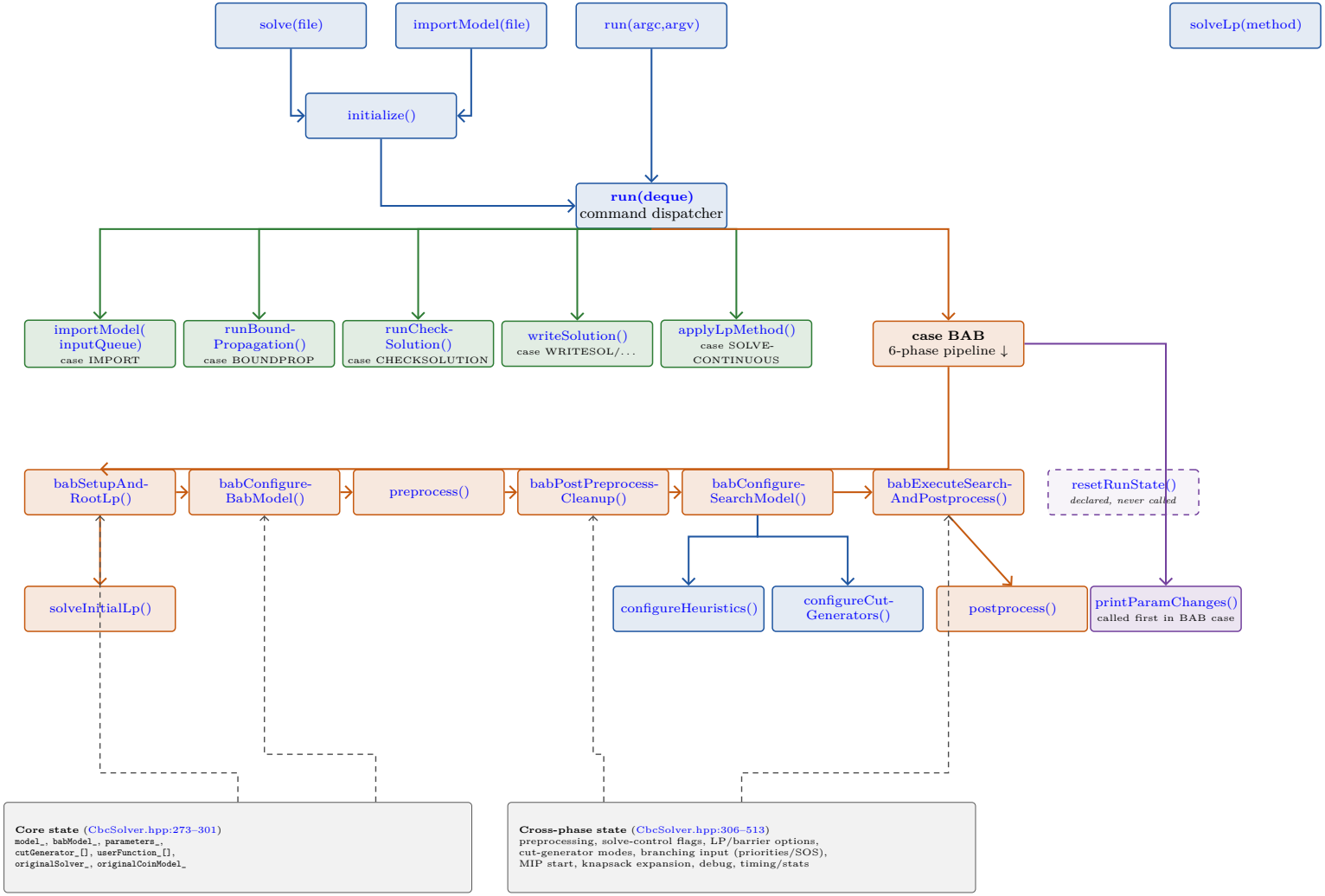
- **IMPORT:** private importModel(inputQueue,...) reads an MPS/LP/GMPL file.
- **SOLVECONTINUOUS:** applyLpMethod() solves the LP relaxation only.
- **BOUNDPROP:** runBoundPropagation() runs standalone bound propagation.
- **CHECKSOLUTION:** runCheckSolution() writes a solution-validation report.
- **WRITESOL / PRINTSOL / ...:** writeSolution() writes solutions in various formats.
- **Bookkeeping:** printParamChanges() prints a parameter-change banner at the top of the BAB case; resetRunState() is declared/defined but currently *never called* (orphaned — dashed outline in the diagram).

4 State organization

Private data falls into two groups. **Core state** (model_, babModel_, parameters_, cutGenerator_, userFunction_, ...) existed in the original class. **Cross-phase state** (lines 306–513) — preprocessing (process_, saveSolver_), solve-control flags, LP/barrier options, cut-generator modes, branching input (priorities/SOS/pseudo-costs), MIP start, knapsack-expansion buffers, debug, lot-sizing, and timing/statistics — used to be *local variables* inside the monolithic run(). Promoting them to member fields is what let run() be split into the cooperating methods of §2: each extracted method reads/writes the subset of this state it needs directly, instead of receiving dozens of by-reference parameters.

Colour legend used on the diagram (next page; node fill and arrow colour share the same code, so each call arrow is coloured like the method it leaves): public API · BAB pipeline (§2) · other run() actions · bookkeeping helpers · state / data structures (dashed = data reads/writes)

CbcSolver — method call graph & state ownership (node labels are hyperlinks to `coin-or/Cbc@next`)



Arrows show call/return relationships within the modularized `BAB` pipeline (§2); dashed grey arrows indicate representative reads/writes of the shared state panel on the left (most pipeline methods touch several state groups — only the busiest links are drawn to keep the diagram readable). All labels link to the corresponding line in `coin-or/Cbc`, branch `next`.